

Artificial intelligence for improving cardiovascular surgery training: A multiyear national exam analysis

Bedirhan Buğra Bayıcı¹, Fatih Kızılyel¹, Hatice Şahin², Yüksel Atay³, Mehmet Kaplan⁴, Murat Sargın⁴,
Murat Bülent Rabuş¹

¹Department of Cardiovascular Surgery, University of Health Sciences Türkiye, Koşuyolu High Specialization Training and Research Hospital, İstanbul, Türkiye

²Department of Medical Education, Ege University Faculty of Medicine, İzmir, Türkiye

³Department of Cardiovascular Surgery, Ege University Faculty of Medicine Hospital, İzmir, Türkiye

⁴Department of Cardiovascular Surgery, University of Health Sciences Türkiye, Dr. Siyami Ersek Thoracic and Cardiovascular Surgery Training and Research Hospital, İstanbul, Türkiye

Abstract

Background: Cardiovascular surgery residency training requires the integration of extensive theoretical knowledge with advanced visuospatial reasoning and technical skills. This study analyzed data from the National Resident Assessment Examinations conducted by the Turkish Society of Cardiovascular Surgery (TSCVS) Board to identify specific training domains in which residents experience the greatest challenges. Furthermore, it explores the potential of artificial intelligence (AI) to support theoretical knowledge acquisition, thereby enabling residents to devote more time to developing visuospatial reasoning and operative decision-making skills.

Methods: In this retrospective observational study, 300 multiple-choice questions from six National Resident Assessment Examinations conducted by the TSCVS Board between 2023 and 2025 were analyzed. Residents were categorized into Junior (13-30 months) and Senior (31-48 months) cohorts according to residency duration. The same questions were answered by the ChatGPT 5.2 model using a zero-shot approach, and AI and resident performances were statistically compared. Questions were also categorized according to topic and cognitive domain, and visual questions were further subclassified.

Results: Residents achieved a mean examination score of 58.1 ± 11.8 , whereas the AI model significantly outperformed the residents, achieving a score of 81.3 ± 6.4 ($p < 0.001$). Junior and Senior residents demonstrated relatively similar mean scores (60.6 vs. 55.6). AI showed high and consistent performance on text-based and guideline-oriented questions; however, its performance declined on examinations requiring visuospatial reasoning and angiographic interpretation. This trend suggests that visually oriented clinical reasoning remains a challenging domain for both human trainees and AI models. AI performance was highest on clinical management and guideline-based questions, whereas lower accuracy was observed on imaging-based and inference-driven questions.

Conclusion: These findings suggest that AI may serve as a valuable educational adjunct for theoretical knowledge acquisition, potentially allowing greater emphasis on simulation-based visuospatial reasoning and operative decision-making skills during residency training.

Keywords: Cardiovascular surgery, residency training, proficiency examination, artificial intelligence, large language models, surgical education.

Cardiovascular surgery residency training is among the most demanding areas of medical education, requiring the integration of extensive theoretical knowledge, advanced visuospatial reasoning, technical skill development, and high-stakes clinical decision-making.^[1] Beyond mastering guideline-based algorithms and evidence-based practices, residents must accurately interpret complex imaging findings and incorporate them into operative strategies. The rapid expansion of medical knowledge, coupled with increasing clinical responsibilities, places a substantial cognitive burden on trainees.^[2]

National resident examinations serve as important tools for the standardized assessment of educational outcomes across training centers.^[3] Although they do not directly evaluate technical skills, they provide reliable measures of theoretical knowledge, clinical reasoning, and problem-solving abilities. The National Resident Assessment Examinations conducted by the Turkish Society of Cardiovascular Surgery (TSCVS) Board offer a structured and comprehensive framework for evaluating residents at different stages of training while supporting national educational standardization.

Cite this article as: Bayıcı BB, Kızılyel F, Şahin H, Atay Y, Kaplan M, Sargın M, et al. Artificial intelligence for improving cardiovascular surgery training: a multiyear national exam analysis. Turk J Thorac Cardiovasc Surg. [Epub Ahead of Print]

Corresponding Author: Murat Bülent Rabuş,

e-mail: muratrabus@yahoo.com **ORCID:** orcid.org/0000-0002-4952-8637

DOI: 10.4274/tjtc.2026.26-2-40

Received: February 20, 2026 **Accepted:** June 23, 2026 **Epub:** July 21, 2026



Copyright© 2026 The Author(s). Published by Galenos Publishing House on behalf of Turkish Society of Cardiovascular Surgery and Turkish Thoracic Surgery Association. This is an open access article under the Creative Commons AttributionNonCommercial 4.0 International (CC BY-NC 4.0) License.

In recent years, artificial intelligence (AI) models have attracted increasing attention in medical education because of their ability to rapidly access, process, and synthesize large volumes of information.^[4] Recent studies have also highlighted the expanding role of AI in cardiovascular medicine, including its applications in clinical decision support and physician education.^[5] Strong performance in theoretical and guideline-based domains has prompted discussion regarding the potential role of AI in specialty training.^[6-8] Rather than serving as a competing entity, AI may be better conceptualized as a supportive educational tool that facilitates access to theoretical knowledge and helps reduce cognitive load within surgical training.

Within this context, evaluating resident performance can provide valuable insight into the domains of cardiovascular surgery training that may benefit from cognitive support and those that remain inherently dependent on visuospatial ability, clinical experience, and surgical judgment.^[9] This distinction is particularly relevant given the central role of imaging interpretation, three-dimensional anatomical understanding, and intraoperative decision-making in cardiovascular surgical practice.

The primary aim of this study was to use multiyear national resident assessment examination data to identify training domains that present the greatest challenges for residents. A secondary aim was to explore whether AI could facilitate theoretical knowledge acquisition, potentially allowing residency programs to place greater emphasis on visuospatial reasoning, operative decision-making, and simulation-based learning.

Methods

This study utilized data obtained from the National Cardiovascular Surgery Resident Assessment Examinations administered by the TSCVS Board between 2023 and 2025. The analysis included a total of 300 multiple-choice questions derived from six examination sessions. All data were de-identified prior to analysis to ensure participant confidentiality. The study was approved by the Local Institutional Ethics Committee of University of Health Sciences Türkiye, Koşuyolu High Specialization Training and Research Hospital (decision date: January 13, 2026; decision no: 2026/01/1332). The requirement for individual informed consent was waived because of the retrospective study design and the use of anonymized data.

Residents were stratified according to their duration of residency training at the time of each examination. The cohort was divided into two groups: The Junior group (13-30 months of training) and the Senior group (31-48 months of training). This stratification was performed to facilitate comparative evaluation of performance patterns across different stages of cardiovascular surgery residency and to examine educational needs associated with training seniority. It should be noted that the National Resident Assessment Examinations are specifically designed for residents within the first 48 months of training. Residents beyond this stage are generally evaluated through a separate board certification examination process with a different structure and participant profile. Therefore, individuals with more than 48 months of training were excluded from the present analysis. A total of 530 examination entries were analyzed across all examination sessions. After duplicate entries were removed, 376 unique residents were identified.

The National Resident Assessment Examinations consist of multiple-choice questions designed to evaluate theoretical knowledge, clinical reasoning, and visual interpretation skills. The question bank includes

guideline-based and text-based clinical scenarios as well as visual materials such as radiographic images, electrocardiographic tracings, and anatomical diagrams. This multimodal structure enables a comprehensive assessment extending beyond factual recall to evaluate complex clinical decision-making and visuospatial competence.

Visual questions were further subclassified into anatomical diagrams/schematics, imaging-based modalities (including chest radiographs and angiographic images), and intraoperative images. AI performance was evaluated descriptively across these categories. In addition, questions were categorized according to their primary topic (anatomy, embryology, physiology, clinical management, and treatment/surgical management) and cognitive type (direct knowledge recall, inference-based reasoning, treatment decision-making/algorithms, and guideline-based questions). AI performance was also evaluated descriptively across these domains.

Question classification according to topic and cognitive type was performed independently by two investigators. Any disagreements were resolved through consensus discussion. Formal assessment of inter-rater agreement (e.g., Cohen's kappa) was not performed.

The ChatGPT 5.2 model (OpenAI, San Francisco, CA, USA), developed using the GPT-5 architecture, was included in the analysis. The AI model generated responses to examination questions using a zero-shot approach. Both text-based questions and questions containing visual content were presented directly to the model, and the resulting responses were evaluated against the official examination answer keys.

All questions were presented to the AI model using a standardized prompt format without additional contextual guidance or iterative refinement. Each question was entered only once, and the first response generated by the model was recorded without modification or regeneration. No feedback, correction, or multiple-attempt strategy was used in order to preserve the integrity of the zero-shot evaluation framework. This approach was selected to reflect real-world single-attempt usage scenarios and to ensure methodological consistency across all questions.

Statistical Analysis

Statistical analyses were performed using Jamovi software (version 2.5; Sydney, Australia). Continuous variables are presented as mean $\pm$ standard deviation, whereas categorical variables are presented as frequencies and percentages. Data distribution was assessed using the Shapiro-Wilk test and Q-Q plots. Independent-samples t-tests and One-Way Analysis of Variance were used for between-group comparisons, as appropriate. Welch's correction was applied when the assumption of homogeneity of variance was violated. Statistical significance was defined as $p < 0.05$.

Results

Overall Performance Distribution and Reference Levels

The analysis included 300 questions from six national examination sessions conducted between 2023 and 2025. In the cumulative analysis of all cohorts, residents achieved a mean score of 58.1 ± 11.8 , whereas the AI model demonstrated significantly superior performance, achieving a mean score of 81.3 ± 6.4 ($p < 0.001$). Notably, the AI model consistently outperformed the mean resident score across all examination sessions analyzed.

Performance Patterns According to Seniority Level

Stratified analysis based on training duration revealed distinct performance patterns at different stages of residency training (Table 1). In assessments representing the early phase of training (Group 1: Junior Residents), the three-year mean resident score was 60.6, whereas the AI model achieved a mean score of 83.3. In examinations representing the advanced phase of training (Group 2: Senior Residents), the mean resident score was 55.6. This finding should be interpreted in the context of differences in examination structure between the junior and senior assessments. In contrast, the AI model maintained a mean score of 79.3 in this cohort, indicating consistently high performance with minimal variation regardless of the intended training level or question complexity.

Effect of Question Modality on Performance

Year-specific analyses highlighted the substantial influence of question modality on performance outcomes (Figure 1). In the 2023 and 2025 examination sessions, which consisted predominantly of text-based and guideline-oriented questions, the AI model consistently achieved scores exceeding 80 across both seniority cohorts. For example, in the 2025 Group 2 examination, the mean resident score was 59.4, whereas the AI model achieved a score of 86.0. Of the 300 questions analyzed, 30 (10.0%)

contained visual components. The distribution of visual questions varied across examination years, with 8 questions in 2023, 17 in 2024, and 5 questions in 2025 (Table 2).

Further subclassification of visual questions revealed considerable variability in AI performance across different visual domains (Table 3). Accuracy was highest for anatomical diagram-based questions (75.0%), whereas substantially lower performance was observed for imaging-based questions (42.9%) and intraoperative images (33.3%). These findings suggest that increasing visual complexity and greater reliance on contextual interpretation are associated with reduced AI performance.

2024 Examination and Performance Inflection

The 2024 examination, characterized by a greater proportion of visually oriented questions, was associated with a notable decline in performance for both residents and the AI model (Figure 2). In this examination session, the mean resident score decreased to 49.8, while AI performance on visually intensive items, particularly at the Senior level, declined to 70.0. This concurrent reduction in performance observed in both groups suggests that questions requiring visual interpretation and three-dimensional reasoning represent a particularly challenging competency domain within the examination framework.

Table 1. Comparative examination performance of resident physicians and the ChatGPT 5.2 model, stratified by examination year and seniority level

Year	Resident group	Resident mean score (mean ± SD)	ChatGPT 5.2 score	Score difference (AI advantage)
2023	Junior residents	59.2±11.8	80.0	+20.8
2023	Senior residents	57.8±12.1	82.0	+24.4
2024	Junior residents	61.9±12.3	88.0	+26.1
2024	Senior residents	49.8±11.5	70.0	+20.2
2025	Junior residents	60.7±12.5	82.0	+21.3
2025	Senior residents	59.4±11.4	86.0	+26.6

SD: Standard deviation; AI: Artificial intelligence.

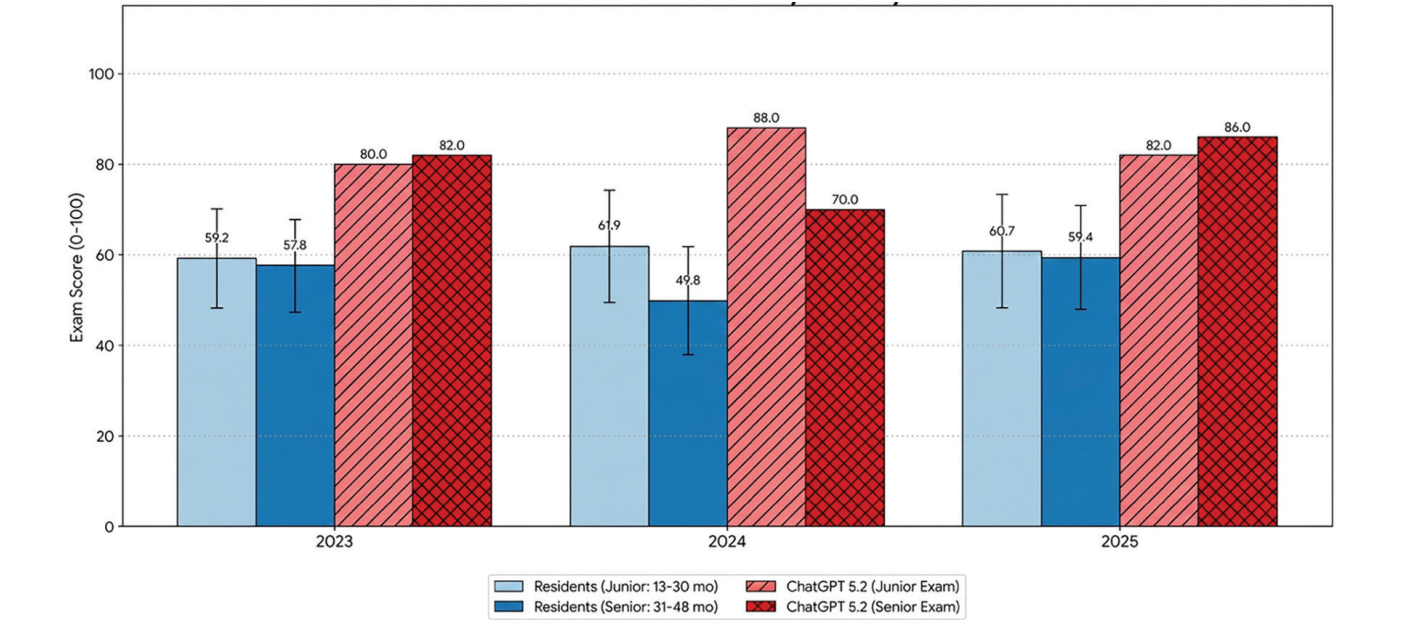


Figure 1. Three-year longitudinal comparative analysis (2023-2025).

Performance Variability and the Effect of Modality

Although the AI model exhibited relatively stable performance on text-based questions, its accuracy declined substantially when confronted with visually oriented content, a pattern that was also observed among residents. Collectively, these findings highlight distinct determinants of performance: Resident success appears to be influenced primarily by training level and clinical complexity, whereas AI performance remains strongly dependent on the modality of the input data.

AI Performance According to Question Topic and Cognitive Type

Analysis of AI performance across content domains revealed variability according to question topic (Table 4). Accuracy was highest for clinical

management (89.3%) and physiology (83.9%) questions, whereas comparatively lower performance was observed in anatomy (72.4%) and treatment/surgical management (76.6%) questions.

Similarly, classification according to cognitive question type demonstrated that the AI model achieved the highest accuracy on guideline-based questions (100.0%) and direct knowledge-recall questions (84.1%). In contrast, lower performance was observed on inference-based reasoning questions (71.0%) (Table 5).

Table 2. Year-wise distribution of visual questions and ChatGPT 5.2 performance (2023-2025)

Year	Correct/total	Accuracy (%)
2023	5/8	62.5
2024	11/17	64.7
2025	3/5	60.0
Overall*	19/30*	63.3*

Table 3. ChatGPT 5.2 performance according to visual question subtype (2023-2025)

Subtype	n	Correct	Accuracy (%)
Anatomical diagrams/schematics	20	15	75.0
Imaging-based (CXR/angiography/other imaging)	7	3	42.9
Intraoperative images	3	1	33.3
Overall*	30*	19*	63.3*

CXR: Chest X-ray.

Table 4. ChatGPT 5.2 performance by topic area (2023-2025)

Topic	n	Correct	Accuracy (%)
Anatomy	76	55	72.4
Embryology	17	14	82.4
Physiology	31	26	83.9
Clinical management	112	100	89.3
Treatment/surgical	64	49	76.6
Total*	300*	244*	81.3*

Table 5. ChatGPT 5.2 performance by question type (2023-2025)

Question type	n	Correct	Accuracy (%)
Direct knowledge recall	182	153	84.1
Inference-based reasoning	69	49	71.0
Treatment decision/algorithm	37	30	81.1
Guideline-based question	12	12	100.0
Total*	300*	244*	81.3*

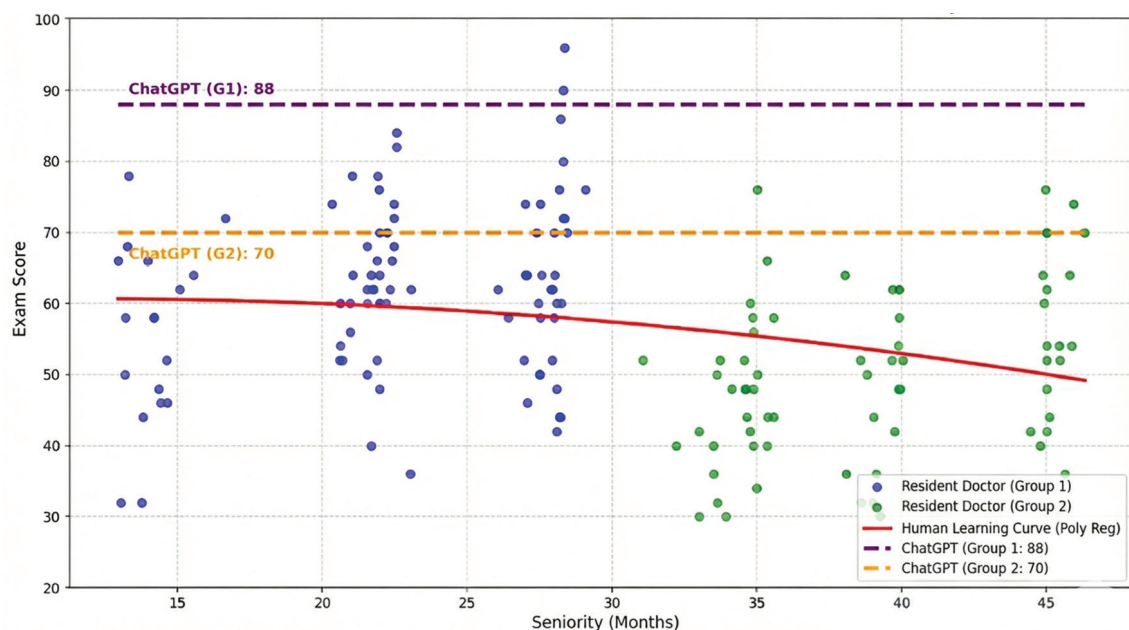


Figure 2. 2024 national resident exam: Resident doctor vs. AI performance analysis.

AI: Artificial intelligence.

Discussion

Rather than reducing the analysis to a simple comparison of superiority, this study provides a performance map that identifies specific domains in which residents encounter the greatest challenges. The observed divergence between AI and resident performance on text-based versus visually intensive questions suggests that theoretical knowledge acquisition and visuospatial surgical reasoning represent distinct cognitive domains with different educational requirements. Within this framework, AI may serve as a supportive educational adjunct that facilitates rapid access to guideline-based knowledge, potentially allowing greater emphasis on experience-dependent competencies such as anatomical orientation, angiographic interpretation, and intraoperative decision-making. These findings support the use of AI not as a replacement for traditional training but as a curricular scaffold for declarative knowledge acquisition, while emphasizing that visuospatial mastery remains an inherently human, experience-dependent competency that requires high-fidelity simulation and operative exposure.

A key finding of this study is the contrast in performance stability between AI and residents. The AI model exhibited minimal performance variation across examinations, suggesting relative independence from examination difficulty or question complexity, whereas resident performance demonstrated a more dynamic pattern that evolved alongside clinical experience and increasing responsibility. AI matched or exceeded resident performance, particularly on guideline-based questions with clearly defined algorithmic responses. This observation suggests that competency levels corresponding to the foundational stages of learning, namely knowledge acquisition and comprehension, may be increasingly amenable to digital support and automation.^[10] However, high performance on theoretical examinations should not be interpreted as evidence of safe or reliable decision-making in real-world cardiovascular surgical practice. In the clinical setting, AI is currently best viewed as a supportive educational tool rather than a substitute for rigorous theoretical training.

The observation that senior residents did not achieve higher examination scores than junior residents should not be interpreted as evidence of limited competency progression. Rather, this finding likely reflects differences in examination structure and difficulty. Senior-level examinations are designed to assess more advanced clinical reasoning, multistep decision-making, and visually intensive interpretation, which may offset the expected performance gains associated with increased training duration.

Additional analyses demonstrated that AI performance was not uniform across content domains. Higher accuracy in clinical management and guideline-based questions suggests strong alignment with structured, algorithm-driven knowledge. In contrast, lower performance in anatomy and inference-based reasoning questions may reflect persistent limitations in spatial understanding and complex contextual reasoning. These findings further highlight the domain-specific strengths and limitations of current AI models in cardiovascular surgical education.

The influence of real-world patient responsibility, complications encountered during on-call duties, and experience gained through the traditional apprenticeship model of surgical training underscores that surgical competence extends beyond the accumulation of theoretical knowledge. This finding supports the concept that clinical competence—

the ability to apply knowledge effectively under conditions of time pressure and uncertainty—is developed primarily through experiential learning.^[11]

The precipitous decline in AI performance during examinations characterized by substantial visual content (e.g., the 2024 session) represents a critical inflection point in our analysis. This performance gap suggests that interpreting angiographic images, computed tomography scans, and three-dimensional spatial relationships requires contextual reasoning that extends beyond simple pattern recognition. For a cardiovascular surgeon, a coronary angiogram is not merely a static image used to quantify stenosis; it is a dynamic source of information that guides revascularization strategy, graft selection, and anticipation of intraoperative challenges. Consistent with this observation, our subtype analysis demonstrated that AI performance varied significantly across visual modalities, with lower accuracy in imaging-based and intraoperative content than in structured anatomical representations. These findings suggest that visual tasks do not impose equivalent cognitive demands and highlight domain-specific limitations of current AI models.

The absence of sensory inputs in AI models—such as haptic feedback (tissue resistance), depth perception, and the dynamic interaction of anatomical planes—may explain why these models struggle with tasks requiring advanced visuospatial cognition.^[9] This finding further underscores the indispensable role of operating room-based practical training, clinical case exposure, and structured visual memory development in cardiovascular surgery education.^[12] In this context, the findings may indirectly suggest that the national assessment examinations administered by the TSCVS Board evaluate distinct competency domains; however, the present study was not designed to formally assess examination validity or psychometric properties.

Furthermore, the analyses revealed that residents exhibited a relatively smaller performance gap compared with AI, particularly on questions requiring complex clinical context, multistep decision-making, and the integration of visuospatial reasoning with clinical judgment. This observation suggests that the assessment instrument may capture not only factual knowledge but also higher-order competencies, including clinical contextualization, visuospatial perception, crisis management, and surgical reasoning.

The results of this study have important implications for the future of surgical education within the framework of Cognitive Load Theory. In an era characterized by the exponential growth of medical literature, expecting residents to retain an ever-expanding body of encyclopedic knowledge is increasingly unrealistic. In this context, the high and consistent performance demonstrated by AI in text-based and guideline-oriented domains suggests its potential value as a strategic support tool for facilitating access to theoretical knowledge and promoting more efficient learning. By enabling rapid access to information and reducing unnecessary information-processing demands, AI may help decrease extraneous cognitive load. This, in turn, may support germane cognitive load by allowing trainees to more effectively organize, process, and integrate information into clinical schemas. However, intrinsic cognitive load, which is associated with the inherent complexity of cardiovascular surgical concepts, three-dimensional anatomical reasoning, and intraoperative decision-making, remains largely dependent on progressive clinical experience and experiential learning.

Residents' critical thinking skills and the educator's confirmatory and instructional roles must remain central to the training process. AI-generated outputs should be regarded as recommendations that support clinical reasoning rather than as definitive decisions. At the same time, concerns regarding the safe and appropriate use of AI in cardiovascular care have received increasing attention, particularly with respect to overreliance and clinical accountability.^[13-15] Furthermore, AI models may generate fabricated reasoning pathways, provide convincing but incorrect responses, or produce hallucinated information that appears plausible despite lacking factual accuracy. In clinical settings, reliance on such outputs without appropriate critical evaluation may result in unsafe recommendations, further underscoring the need for human oversight and expert validation.

Practical Integration of AI into Residency Training

In practical terms, AI systems may be incorporated into cardiovascular surgery training as structured adjunctive tools rather than as standalone decision-makers. Potential applications include AI-assisted guideline retrieval, case-based rehearsal using simulated clinical scenarios, and structured feedback on clinical reasoning processes. In addition, AI may support preoperative planning discussions by summarizing evidence-based strategies and highlighting alternative management approaches. When integrated into supervised educational frameworks, these tools may help optimize the acquisition of theoretical knowledge while preserving the central roles of experiential learning, surgical judgment, and mentor-guided training.

Educational Implications for Residency Curriculum

The concurrent decline in both resident and AI performance during visually intensive examinations underscores that visuospatial mastery remains a critically experience-dependent competency in cardiovascular surgery. Consequently, these findings support a strategic reconfiguration of early residency training, which has traditionally emphasized theoretical knowledge acquisition. By leveraging AI to facilitate access to foundational knowledge, limited training time may be more effectively reallocated to high-fidelity operative simulation, case-based visual reasoning exercises, and supervised intraoperative decision-making. Such an approach may promote the development of essential surgical orientation and visuospatial skills.

This study has several limitations. First, the analysis evaluated a single general-purpose large language model; medically fine-tuned architectures and multimodal models were not assessed. As a result, the findings may not be fully generalizable to specialized AI systems with enhanced capabilities in visuospatial tasks. Second, the static visual stimuli used in board examinations do not fully replicate the dynamic, high-fidelity environment of the operating room, where visual information is integrated with tactile feedback. Because current AI models cannot perceive haptic information, this dimension falls outside the scope of the present comparative analysis. Although current residency training addresses this limitation through operating room-based experience and real-time feedback, the integration of AI into these dynamic, experience-based learning processes remains unclear and was therefore beyond the scope of this study. Despite these limitations, the use of a multiyear, seniority-stratified national dataset strengthens the external validity and generalizability of our findings. Residents with more than 48 months of training were not included because they are evaluated through a

separate board examination system, limiting assessment across the full continuum of residency training. Furthermore, although duplicate entries were identified and accounted for, some residents participated in multiple examination sessions across different years; therefore, a degree of repetition bias cannot be entirely excluded. Additionally, question difficulty was not formally assessed using standardized psychometric measures, such as difficulty and discrimination indices, and detailed subspecialty-based classifications were not performed. These factors may limit the interpretation of performance differences between junior and senior residents. Moreover, although question classification was independently conducted by two investigators, formal inter-rater reliability analysis was not performed, potentially introducing subjectivity into the categorization process. These limitations primarily reflect the absence of item-level resident response data. Future studies incorporating such data may enable more detailed psychometric evaluation and subgroup analyses.

In conclusion, our findings indicate that although residents experience greater difficulty with visually intensive and spatially oriented assessments, current AI models lack the contextual understanding and sensory capabilities necessary to perform effectively in these domains. Conversely, AI may serve as a valuable tool for facilitating access to guideline-based theoretical knowledge, thereby enabling greater educational emphasis on experience-dependent competencies, including mentor-guided training, simulation-based visuospatial skill development, and operative decision-making.

Ethics

Ethics Committee Approval: The study was approved by the Local Institutional Ethics Committee of University of Health Sciences Türkiye, Koşuyolu High Specialization Training and Research Hospital (decision date: January 13, 2026; decision no: 2026/01/1332).

Informed Consent: The requirement for individual informed consent was waived because of the retrospective study design and the use of anonymized data.

Acknowledgments

During the preparation of this work, the authors used ChatGPT 5.2 (OpenAI) in order to improve the language and readability of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Footnotes

Authorship Contributions

Concept: B.B.B., F.K., H.Ş., Y.A., M.K., M.S., M.B.R.; Design: B.B.B., F.K., Y.A., M.K., M.B.R.; Data Collection or Processing: B.B.B., F.K., M.B.R.; Analysis or Interpretation: B.B.B., F.K., H.Ş., Y.A., M.K., M.S., M.B.R.; Literature Search: B.B.B., F.K., M.B.R.; Writing: B.B.B., F.K., H.Ş., M.B.R.

Conflict of Interest: No conflict of interest was declared by the authors.

Financial Disclosure: The authors declared that this study received no financial support.

References

1. Vaporciyan AA, Yang SC, Baker CJ, Fann JJ, Verrier ED. Cardiothoracic surgery residency training: past, present, and future. *J Thorac Cardiovasc Surg.* 2013;146:759-67. doi:10.1016/j.jtcvs.2013.06.004.
2. Young JQ, Van Merriënboer J, Durning S, Ten Cate O. Cognitive load theory: implications for medical education: AMEE guide no. 86. *Med Teach.* 2014;36:371-84. doi:10.3109/0142159X.2014.889290.
3. Epstein RM. Assessment in medical education. *N Engl J Med.* 2007;356:387-96. doi:10.1056/NEJMr054784.
4. Sallam M. ChatGPT utility in healthcare education, research, and practice: systematic review on the promising perspectives and valid concerns. *Healthcare (Basel).* 2023;11:887. doi:10.3390/healthcare11060887.
5. Gocer H, Durukan AB. The use of artificial intelligence in interventional cardiology. *Turk Gogus Kalp Damar Cerrahisi Derg.* 2023;31:420-1. doi:10.5606/tgkdc.dergisi.2023.24791.
6. Kung TH, Cheatham M, Medenilla A, Sillos C, De Leon L, Elepaño C, et al. Performance of ChatGPT on USMLE: potential for AI-assisted medical education using large language models. *PLOS Digit Health.* 2023;2:e0000198. doi:10.1371/journal.pdig.0000198.
7. Huang Y, Goma A, Semrau S, Haderlein M, Lettmaier S, Weissmann T, et al. Benchmarking ChatGPT-4 on a radiation oncology in-training exam and Red Journal Gray Zone cases: potentials and challenges for AI-assisted medical education and decision making in radiation oncology. *Front Oncol.* 2023;13:1265024. doi:10.3389/fonc.2023.1265024.
8. Liu M, Okuhara T, Chang X, Shirabe R, Nishiie Y, Okada H, et al. Performance of ChatGPT across different versions in medical licensing examinations worldwide: systematic review and meta-analysis. *J Med Internet Res.* 2024;26:e60807. doi:10.2196/60807.
9. Hashimoto DA, Rosman G, Rus D, Meireles OR. Artificial intelligence in surgery: promises and perils. *Ann Surg.* 2018;268:70-6. doi:10.1097/SLA.0000000000002693.
10. Salvador R, Vas D, Oleaga L, Matute-González M, Castillo-Fortuño À, Setoain X, et al. Comparing ChatGPT and medical student performance in a real image-based radiology and applied physics in medicine exam. *Radiologia (Engl Ed).* 2025;67:101638. doi:10.1016/j.rxeng.2025.101638.
11. Moulton CA, Regehr G, Lingard L, Merritt C, Macrae H. Slowing down when you should: initiators and influences of the transition from the routine to the effortful. *J Gastrointest Surg.* 2010;14:1019-26. doi:10.1007/s11605-010-1178-y.
12. Petrosioniak A, Lu M, Gray S, Hicks C, Sherbino J, McGowan M, et al. Perfecting practice: a protocol for assessing simulation-based mastery learning and deliberate practice versus self-guided practice for bougie-assisted cricothyroidotomy performance. *BMC Med Educ.* 2019;19:100. doi:10.1186/s12909-019-1537-7.
13. Johnston SC. Anticipating and training the physician of the future: the importance of caring in an age of artificial intelligence. *Acad Med.* 2018;93:1105-6. doi:10.1097/ACM.0000000000002175.
14. Wartman SA, Combs CD. Medical education must move from the information age to the age of artificial intelligence. *Acad Med.* 2018;93:1107-9. doi:10.1097/ACM.0000000000002044.
15. Gocer H, Durukan AB. The potential risks of artificial intelligence in cardiovascular care. *Turk Gogus Kalp Damar Cerrahisi Derg.* 2025;33:262-4. doi:10.5606/tgkdc.dergisi.2025.26912.